

SYED MAOZ ARIF

Hardware Developer & Embedded Systems Engineer

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SUMMARY

Detail-oriented Hardware Developer and Electronics Engineer with a proven track record in multi-layer PCB design, embedded systems, and IoT architecture. Developed 12+ real-world engineering projects and achieved 17 technical certifications. Recognized for innovative problem-solving, securing 1st Position in national engineering exhibitions and a Global Instructables Contest win. Adept at cross-functional leadership, managing 50+ members as IET Chapter President, and delivering high-quality freelance engineering solutions with a 100% client satisfaction rate.

SKILLS

- **Hardware & PCB Design:** Altium Designer, KiCad, Proteus, LTspice, Multisim, Schematic Capture, Impedance Matching, Signal Integrity, 3D CAD (SolidWorks, Fusion 360, AutoCAD)
- **Embedded Systems & IoT:** ESP32, STM32, Arduino, Lattice FPGA, I2C, SPI, UART, RF, MQTT, Firebase
- **Programming Languages:** C/C++ (Embedded C), Python, Verilog (HDL), MATLAB, JavaScript, HTML5/CSS3
- **Core Competencies:** Leadership, Project Scoping, Technical Documentation, Problem Solving, Client Relations, Agile Methodologies

WORK EXPERIENCE

Freelance Hardware & Mechanical Designer

Jan 2024 – Present

Fiverr & Upwork

Remote

- Delivered 10+ professional multi-layer PCB designs, schematic captures, and 3D CAD models for international clients, maintaining a 100% positive rating.
- Managed end-to-end project scoping, component selection (BOM), and deadline management, reducing design iterations by 30%.

President (Previously VP & Membership Head)

May 2025 – Present

IET On Campus, CUI Abbottabad

Abbottabad, PK

- Directed chapter operations and represented the student branch, fostering engineering excellence for 50+ active members.
- Organized 5+ major technical events and specialized workshops (e.g., AI in UAVs), increasing student engagement by 40%.

Cabinet Member

Jan 2025 – Present

Youth Parliament Pakistan

National

- Represented the youth engineering sector in national policy discussions, collaborating with 100+ national delegates on youth advocacy.

PROJECTS

Industrial Edge AI Condition Monitoring & 3D Digital Twin

Jul 2026

- Designed an industrial-grade development board with a dual-MCU architecture (STM32H7, ESP32), integrating 70 GPIOs to monitor 3-phase induction motors.
- Engineered real-time data acquisition systems, reducing system diagnostic time by 40% and securing 1st Position (Software Category) at CEP Exhibition SP26 and 2nd Position at EMCOT 2026.

Stratum-1 Precision Hardware Clock & GPS Bridge

Aug 2026

- Engineered a Stratum-1 time synchronizer using pure digital logic on a Lattice iCE40UP5K FPGA, achieving 0ms operating system latency and 100% synchronization accuracy.
- Developed a custom hardware UART receiver and NMEA state machine in Verilog to parse GPS signals directly in silicon.

ElectroParts IMS: Inventory Architecture

Aug 2026

- Architected a cross-platform inventory management system (CapacitorJS, Firebase) supporting 32 distinct electronics categories, increasing component retrieval efficiency by 50%.
- Implemented dual-storage local/cloud sync and linear barcode generation algorithms to ensure zero-latency data access.

IncliSense Precision Digital Level

Apr 2026

- Built a precision leveling device analyzing dual-axis spatial data via I2C, yielding high-accuracy PC telemetry for real-time monitoring.
- Awarded Global Runner-Up in the Instructables Sensors Contest out of hundreds of international submissions.

EDUCATION

B.S. Electrical Engineering (Electronics)

Expected 2028

COMSATS University Islamabad, Abbottabad Campus

GPA: 3.61/4.0

- **Honors:** Altium Global Scholarship Awardee (March 2026).
- **Focus:** Embedded Systems, PCB Design, Signal Integrity, IoT, Firmware Engineering, Digital Logic.

CERTIFICATIONS

- Altium Certified: Designer Essentials (Altium Education)
- Generative AI Mastermind (Outskill)
- Arduino & IoT Certification (Cardiff Institute)
- 3D Printing & CAD Engineering Certificate
- 13 additional professional certifications in Leadership, Project Management, and Electronics Design.